Exercises 2

Exercise 1 (Bra-ket notation)**.** *Let A be an $n \times n$ matrix given in the computational basis and let $\langle i|$ denote the i-th standard basis row vector and $|j\rangle$ denote the j-th standard basis column vector. Show that*

1. *$\langle i|A$ is the i-th row of A,*
2. *$A|j\rangle$ is the j-th column of A,*
3. *$\langle i|A|j\rangle = A_{ij}$, the (i,j)-th entry of A.*

Let $A = \sum_{k=1}^{n} |\psi_k\rangle\langle k|$, for some vectors $|\psi_k\rangle \in \mathbb{C}^n$. In terms of these vectors,

4. *what is $A|j\rangle$ equal to,*
5. *what is $\langle i|A|j\rangle$ equal to?*

Exercise 2 (Properties of the inner product)**.** *Verify that the inner product $\langle u|v\rangle$ on $\mathbb{C}^n$ satisfies the following two properties:*

1. *$\langle u|v\rangle = \big(\langle v|u\rangle\big)^\dagger = \overline{\langle v|u\rangle}$,*
2. *$\langle u|u\rangle \geq 0$ and $\langle u|u\rangle = 0$ iff $|u\rangle = \mathbf{0}$ (the all-zeroes vector).*

Exercise 3 (Properties of matrix multiplication)**.** *The goal of this exercise is to verify some basic properties of matrix multiplication.*

1. *Let $A = |i\rangle\langle j|$, $B = |k\rangle\langle l|$, $C = |s\rangle\langle t|$, for some $i,j,k,l,s,t \in \{1,\ldots,n\}$, be three standard basis matrices. Verify that*

$$(AB)C = A(BC).$$

   Why does this imply the same equation for all complex matrices $A, B, C \in \mathrm{M}_{n,n}(\mathbb{C})$?

2. *Let $A = |i\rangle\langle j|$ and $B = |k\rangle\langle l|$, for some $i,j,k,l \in \{1,\ldots,n\}$, be two standard basis matrices. Verify that*

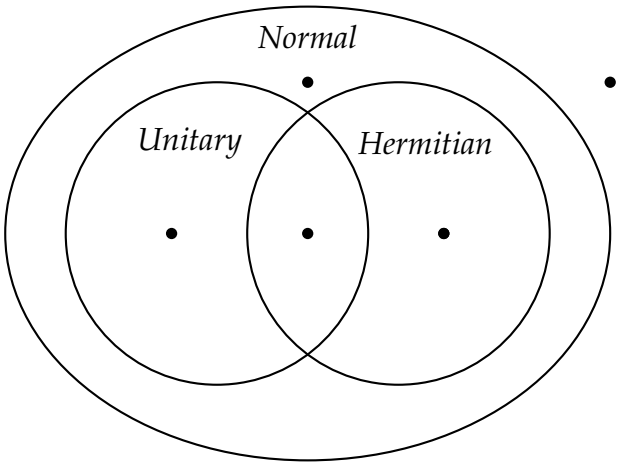
$$(AB)^\mathsf{T} = B^\mathsf{T} A^\mathsf{T}.$$

   Why does this imply the same equation for all complex matrices $A, B \in \mathrm{M}_{n,n}(\mathbb{C})$?

Exercise 4 (Types of matrices). *Consider the following five matrices:*

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}, \quad D = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad E = \begin{pmatrix} 2 & 0 \\ 0 & i \end{pmatrix}.$$

Match them with the five points in this diagram:



Exercise 5 (Diagonalizing Y). *The goal of this exercise is to diagonalize* $Y = \left(\begin{smallmatrix} 0 & -i \\ i & 0 \end{smallmatrix}\right)$.

1. *Find the eigenvalues* λ_1, λ_2 *of* Y.
2. *Find corresponding eigenvectors* $|v_1\rangle$, $|v_2\rangle$ *(of unit length) of* Y.
3. *Verify that* $Y = \lambda_1 |v_1\rangle\langle v_1| + \lambda_2 |v_2\rangle\langle v_2|$.

Exercise 6 (Eigenvalues, eigenvectors, and diagonalisation). *Let* $A \in \mathrm{M}_{n,n}(\mathbb{C})$ *be a normal matrix and* $A = \sum_k \lambda_k |v_k\rangle\langle v_k|$ *be its diagonal form (also known as "spectral decomposition" of A). Recall that here* λ_k *are the eigenvalues of* A *and* $|v_k\rangle$ *are the eigenvectors of* A*, and that the eigenvectors are orthonormal:* $\langle v_i | v_j \rangle = \delta_{ij}$.

1. *Show that* $\lambda_k \in \mathbb{R}$ *for all* k *if and only if* A *is Hermitian and that* $|\lambda_k| = 1$ *for all* k *if and only if* A *is unitary.*
2. *What is the spectral decomposition of* $A^2 \equiv A \cdot A$*?*
3. *Show that the matrix* $\exp(A) \equiv \sum_{n=0}^{\infty} A^n / n!$ *has spectral decomposition* $\sum_k e^{\lambda_k} |u_k\rangle\langle u_k|$. *Show that* $\exp(-iA)$ *is unitary if* A *is Hermitian.*
4. *Assume* A *is Hermitian* and *unitary. What are the eigenvalues of* A*? How does* A^2 *look like?*

Exercise 7 (Tensor product). *Let* $I = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ *and* $M = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. *Compute explicitly the* 4×4 *matrices* $I \otimes M$ *and* $M \otimes I$.